

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

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The Transformer Architecture, introduced in the original paper[2], is a type of neural network originally designed for natural language processing (NLP) tasks such as translation or text generation. Unlike the earlier models like Recurrent Neural Network (RNNs) or Long Short-Term Memory (LSTMs), which process data sequentially with each word depending on the previous one, Transformers do not process data step by step. Instead, they use a mechanism called self-attention to understand the relationships between all parts of a sequence simultaneously. This very aspect of the architecture allows the model to capture long-range dependencies efficiently and makes it highly parallelizable by letting many parts of the model's computation to happen at the same time.

In the paper[2], the author introduced Vision Transformer (ViT), which adapts the aforementioned architecture with the intention of language, directly to image classification. Instead of using Convolutional Neural Networks (CNNs), ViT divided an image into small fixed-size patches, converts each patch into a vector embedding (a flattened, numerical representation that captures the features or information of that patch), and feeds the sequence of these embeddings into a standard Transformer encoder. During this process, a learnable [CLS] token (a special token that gathers information from all patches and represents the overall meaning of the entire image for classification) is also added with the embeddings.

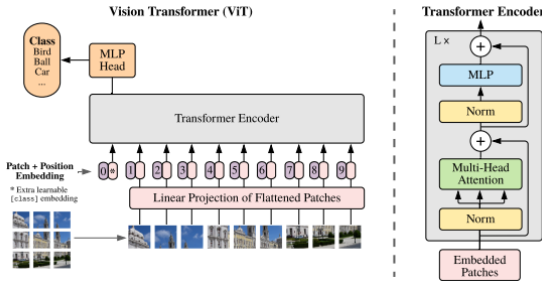


Figure 1: Overview of the Vision Transformer (ViT) model. The image is divided into fixed-size patches, which are embedded and passed through a Transformer encoder[8] along with a learnable [CLS] token for classification.

In details provided by the author, it was learnt that ViT follows the same architecture as the original Transformer (Vaswani et al., 2017)[8], but applies it directly to images instead of text. This intentionally simple setup allows researchers to reuse existing, efficient, and scalable NLP Transformer implementations.

Firstly, an image of size $(H \times W \times C)$ is divided into small fixed-size patches of size $(P \times P)$. Afterwards, each patch is flattened and linearly projected into a D-dimensional vector (embedding). The total number of patches, $N = (H \times W) / P^2$, forms the sequence length for the Transformer. Also, a learnable [CLS] token is prepended to the sequence of patch embeddings. Here in this token, a classification head (MLP during pre-training, linear layer during fine-tuning) is attached. After passing through all Transformer layers, the [CLS] token's output represents the whole image and is used for classification. Furthermore, the authors said that since Transformers don't understand spatial order, both standard learnable 1D positional embeddings and more advanced 2D-aware position embeddings were added to each patch embedding for testing. Upon testing, it was learnt by the authors that the latter showed no significant performance gains over the former.

The Transformer encoder is composed of L alternating layers of: Multi-Head Self-Attention (MSA), which enables each patch to attend to all other

patches, and MLP (Feedforward Network) — processes information from the attention output. Each block uses Layer Normalization (LN) before the operation and residual connections after it to stabilize training and help gradient flow. The MLP contains two fully connected layers with a GELU activation in between.

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}}, \quad \mathbf{E} \in R^{(P^2 \cdot C) \times D}, \quad \mathbf{E}_{\text{pos}} \in R^{(N+1) \times D} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L \quad (2)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L \quad (3)$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \quad (4)$$

The above equations (1)–(4) describe the process:

(1) Input sequence = [CLS] token + patch embeddings + positional embeddings.

(2) Apply self-attention with residual connection.

(3) Apply MLP with residual connection.

(4) The final image representation is obtained from the normalized output of the [CLS] token after the last layer.

Moreover, ViT has far less built-in image-specific inductive bias than CNNs, which inherently capture locality and translation equivariance. ViT uses 2D structure only when dividing images into patches and adjusting positional embeddings, meaning it must learn spatial relationships entirely from data. Additionally, a hybrid version can use CNN feature maps as input, combining CNN's local feature extraction with the Transformer's global attention. During fine-tuning, the Vision Transformer (ViT) replaces its pre-trained classification head with a new layer that matches the number of target classes. It is often fine-tuned on higher-resolution images while keeping the patch size the same, which increases the sequence length. To handle this, ViT uses 2D interpolation to adjust positional embeddings, which is the only stage where explicit 2D spatial information is manually introduced.

The authors also evaluated ViT against ResNet and hybrid CNN–Transformer models to compare performance, data requirements, and computational cost. ViT was pre-trained on datasets of increasing size (ImageNet (1.3M) > ImageNet-21k (14M)[1] > JFT-300M (303M)[7]) and tested on multiple benchmarks such as CIFAR-10/100[4], Oxford Flowers[5], Oxford-IIIT Pets[6], and VTAB (19 tasks). Three ViT variants were used: Base (86M params), Large (307M), and Huge (632M). Training used Adam optimizer, large batch sizes, and high weight decay, followed by SGD fine-tuning at higher image resolutions. Upon analyzing the results, it was observed by the authors that ViT models outperformed CNNs like ResNet (BiT) and EfficientNet on nearly all benchmarks when pre-trained on large datasets, while requiring 2–4× less compute. Also, ViT-H/14 achieved 88.55% on ImageNet, 94.55% on CIFAR-100, and 77.63% on VTAB, surpassing state-of-the-art models. With smaller datasets, CNNs performed better due to their inductive biases, but ViT excelled as data size increased, proving that large-scale training compensates for lack of built-in visual bias.

Furthermore, the scaling study showed that Vision Transformers (ViT) outperform ResNets[3] in accuracy while using 2–4× less computation, and their performance continues to improve as model size increases. Hybrid models combining CNN and ViT perform slightly better at small scales but lose this advantage with larger models. The analysis of ViT's internal behavior revealed that it learns both local and global dependencies, where attention heads focus on either small regions or the entire image. Finally, a self-supervised experiment using masked patch prediction improved ViT's performance by 2%, showing promise for future unsupervised learning approaches.

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